TowardsHolisticEmotionUnderstanding:A Text-AnchoredCross-AttentionTransformerfor MultimodalSentimentAnalysis

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***Abstract*— Multimodalsentimentanalysisrequiresintegrating textual,acoustic,andvisualcuestoaccuratelyinterprethu-manaffect.Thispaperinvestigateswhetheracross-attention Transformer,anchoredtothetextmodality,canbetterfuse acousticandvisualinformationthanconventionalearlyorlate fusionstrategies.Weevaluateallthreeapproacheson CMU-MOSEIthroughelevenexperimentscoveringbaselinecom-parisons,modalityablations,andhyperparametersweeps.The proposedText-AnchoredCross-AttentionTransformer,trained with16attentionheadsanda128-dimensionalembeddingspace, achieves80.03%binaryaccuracyandaweightedF1of0.8687— outperformingearlyfusionat78.14%andlatefusionat77.23%. Akeyfindingisthatcross-attentiononlyoutperformssimpler baselinesoncesufficientrepresentationalcapacityisprovided; afour-headvariantactuallyunderperformedbothbaselines. Attentionheatmapsandt-SNEprojectionsconfirmthatthe model learns to anchor on emotionally salient frames rather than trackingword-leveltemporalcorrespondence.**

***IndexTerms*—multimodalsentimentanalysis,cross-attention transformer,emotionrecognition,text-anchoredfusion,CMU-MOSEI**

I.INTRODUCTION

"I'mfine"deliveredwithadownwardgazeandafalling

SENTIMENTisnotjustaboutwhatsomeonesays.Aflat pitch carries very different meaning from the same phrase said withasmile.Modelsthatreadonlythetranscriptwillget thiswrongeverytime.Thisfoundationalconceptinaffective computing[1]hasdrivenamassiveshifttowardmultimodal analysis[2],[3].

While this integration is conceptually straightforward, build-ingasystemthateffectivelyutilizesallthreechannels— text, audio,andfacialexpression— presentssignificantchallenges comparedtosimplyfeedingallthreeintoasingle model.The fusionstrategymattersenormously.Concatenatingfeatures earlyonmixescleanandnoisyinformationinwaysthatcon-fusethemodel.Trainingthreeseparatemodelsandaveraging theiroutputsavoidsthatproblem,butlosesanysenseofhow thechannelsinteractwitheachother.Theinterestingquestion is whether an attention mechanism can do better— specifically, whether the model can learn to query audio and visual streams throughthelensofwhatisbeingsaid.

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Existing text-anchored methods primarily focus on architectural design. In contrast, this work systematically investigates the effect of representational capacity on cross-attention performance through controlled experiments, ablation studies, and qualitative attention analysis.

We propose a Text-Anchored Cross-Attention Transformer in which text serves as the query modality in two parallel cross-attention layers attending to audio and visual features respectively. The architecture was trained and evaluated on CMU-MOSEI through a three-phase experimental study producing eleven distinct results. Implementation was done in PyTorch on an NVIDIA RTX 3050 Laptop GPU.

This paper makes three contributions. First, a practical cross-attention fusion architecture is proposed and evaluated against early and late fusion baselines under identical training conditions. Second, a systematic ablation study quantifies the independent contribution of each modality under the proposed framework. Third, qualitative analysis through attention heatmaps, alignment maps, and embedding space visualizations provides insight into what the model actually learned.

II. Literature Review

A. Multimodal Sentiment Analysis

The field was largely shaped by the release of CMU-MOSI [4], which gave researchers a standardized benchmark for sentiment prediction from video clips. Accuracy improvements since then have come mostly from two directions: better text representations and better fusion strategies [2], [3].

On the text side, the shift from GloVe [5] to BERT [6] was significant. GloVe assigns a single fixed vector to each word regardless of context. BERT produces different representations for the same word depending on the sentence it appears in, which makes a real difference for sentiment where the same word can be positive or ironic depending on surrounding context. Most published systems achieving above 82% on CMU-MOSEI use BERT or RoBERTa. This work uses GloVe because those are the pre-extracted features available in the CMU-MOSEI SDK, and extracting BERT features from scratch was outside the scope of this phase.

CMU-MOSEI [7], used as the evaluation dataset in this work, extends CMU-MOSI to 23,453 clips from over 1,000 speakers and adds six-class emotion labels alongside the sentiment regression task. It is the largest available benchmark for this problem.

B. Fusion Approaches

Early fusion concatenates all modality features before the model sees them. Zadeh et al. [8] showed that the full outer-product Tensor Fusion Network becomes computationally intractable at scale. Liu et al. [9] proposed low-rank factorisation as a more practical approximation. Even so, the core problem with early fusion is that noisy features from one modality corrupt the joint representation.

Late fusion avoids this by keeping modalities separate until the final prediction step. It is simpler to train and more robust to missing inputs. But it cannot detect the interaction between a specific word and the facial expression occurring at the same moment—information that is only visible if the two streams are compared at the feature level.

Cross-attention addresses both problems. Tsai et al. [10] introduced the Multimodal Transformer (MulT), which applies directed cross-attention across all modality pairs and consistently outperformed LSTM-based baselines on CMU-MOSI and CMU-MOSEI. Hazarika et al. [11] later proposed MISA, which disentangles modality-specific and modality-shared representations through subspace projection, achieving strong results on both datasets. More recent architectures add gating mechanisms to suppress noisy modality contributions [12], graph structures to capture conversational turn dynamics [13], or dual-level feature restoration [14], [15]. Other notable works include emotion-cause extraction [16] and multi-task hierarchical fusion [17].

The general trend is clear: cross-attention-based approaches [18], [19], [20], [21] outperform simpler fusion strategies, and text-oriented designs where text provides the query [22] tend to work better than symmetrical alternatives.

Existing methods can broadly be grouped into three categories: early fusion strategies that concatenate features at the input level, late fusion approaches that independently process modalities before final aggregation, and cross-attention architectures that dynamically align representations across modalities. While early and late fusion offer simplicity and computational efficiency, cross-attention methods consistently demonstrate superior capacity for capturing complex inter-modal dependencies, particularly when text is utilized as an anchoring modality.

C. Summary Table

TABLE I
SELECTED LITERATURE ON MULTIMODAL FUSION

Method	Strength	Weakness	Dataset
MulT [10]	Long-range cross-modal dependencie	High memory for long sequences	CMU-MOSI, CMU-MOSEI, IEMOCAP
MISA [11]	Disentangles shared and specific features	Fails when modalities are missing	CMU-MOSI, CMU-MOSEI
TCAN [22]	Text pivot guides noisy audio/visual streams	Degrades with poor transcription	CMU-MOSI, CMU-MOSEI, IEMOCAP
GA2MIF [13]	Captures temporal and inter-modal patterns	Adds latency	IEMOCAP, MELD
GCMA-Net [12]	Suppresses noisy modality signals	Sensitive to domain shifts	CH-SIMS

D. Datasets Overview

TABLE II
BENCHMARK DATASETS

Dataset	Size	Text	Audio	Vision
CMU-MOSI [4]	2,199 clips	GloVe	COVAREP	FACET
CMU-MOSEI [7]	23,453 clips	GloVe	COVAREP	FACET
IEMOCAP [23]	9,112 utts.	Raw	.wav	.avi
MER2024 [24]	5,030 clips	Subtitles	Speech	Video

III. METHODOLOGY

A. Problem Statement

An utterance is represented by text features $T \in \mathbb{R}^{L \times 300}$, audio features $A \in \mathbb{R}^{L \times 74}$ and visual features $V \in \mathbb{R}^{L \times 35}$, where $L = 50$ is the padded sequence length. The objective is to learn a mapping $F(T, A, V) \rightarrow y$ where y is either a continuous sentiment score for regression or a binary label for classification.

B. Proposed Architecture

The architecture is motivated by a simple observation: text is the most structured of the three modalities. It is already tokenized and semantically dense, making it a reliable anchor. Rather than having text, audio, and vision contribute symmetri-cally, the design uses text as the query modality in two parallel cross-attention layers—one attending to audio, one attending to vision.

Modality Projection. Each input stream is projected into a shared 128-dimensional embedding space via learned linear layers. The projection parameters are trained end-to-end with the rest of the model.

Layer Normalisation. Applied to each projected representation before cross-attention. In practice, skipping this step caused validation accuracy to oscillate significantly in early training epochs—adding normalisation stabilised convergence noticeably.

Cross-Attention (Text→Audio). Text provides Query vectors Q; audio provides Key and Value vectors K and V :

$$\text{Attention}(Q, K, V) = \text{softmax}\frac{QK_T\sqrt{d_k}}{V} \quad (1)$$

where $d_k = 128/16 = 8$ per head. The output is a text-conditioned audio representation where each word position has attended to the most relevant acoustic frames.

Cross-Attention (Text→Vision). Identical mechanism, with visual features as Key and Value. This runs in parallel with the audio cross-attention, not in sequence.

Residual Connections. The original text representation is added back to each cross-attention output before the concatenation step. This follows standard Transformer practice and

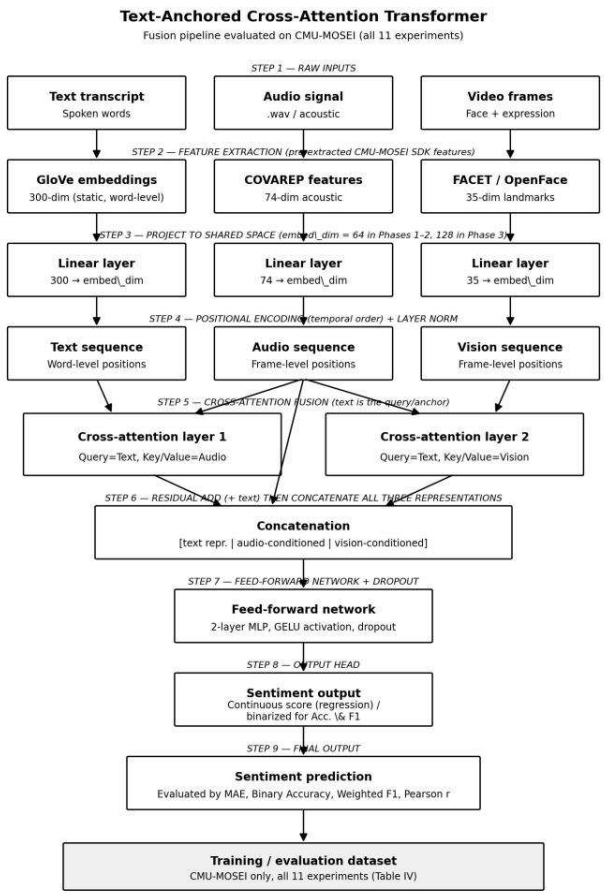


Fig.1.TheproposedText-AnchoredCross-AttentionTransformer.Textactsasthequerymodalityintwoparallelcross-attentionlayers,eachattendingtooneoftheothermodalities.

helpedgradientflowduringbackpropagation,particularlyinlongertrainingruns.

ConcatenationandClassification. Allthreerepresentationsareconcatenatedandpassedthroughatwo-layerfeed-forwardnetworkwithGELUactivationanddropout.Afinallinearlayerproducesascalarsentimentscore.

Multi-headattentionwith16parallelheadsisusedinthefinal model. The significance of this choice is discussed in the resultssection.

C.Baselines

EarlyFusion. Text(300-dim),audio(74-dim),andvisual(35-dim)featuresareconcatenatedintoa409-dimensionalvectorandprocessedthroughatwo-layerLSTM.Simpletoimplement;servesasthelower-boundreference.

LateFusion. Threeindependent LSTMnetworksprocesseachmodalityseparately.Theiroutputsentimentscoresarecombinedviaweightedaveraging.Testswhetherindependent

after verifying that larger batches caused out-of-memory errors duringthefullthree-modalitycross-attentionforwardpass. The AdamW optimizer was used with weight decay $1e{-}4$ and gradientclippingatmaxnorm=1.0—thelatterwasimpor-tant because without it, the cross-attention models occasionally produced gradient spikes in the first few epochs. Learning rate scheduling used ReduceLROnPlateau (factor=0.5, patience=5). Earlystoppingwassettopatience=10epochsbasedonval-idationaccuracyratherthanvalidationloss;thisdistinctionturnedouttomatter,asthetwometricsdidnotalwaysagreeonwhichcheckpointwasbest.

TABLEIII
TRAINING CONFIGURATION

Parameter	Value
Optimizer	AdamW(weightdecay=1e-4)
LearningRate	5e-4(Phase1&2);3e-4(Phase3)
LossFunction	L1Loss(MeanAbsoluteError)
BatchSize	32
Epochs	10(Phase1&2);20(Phase3)
GradientClipping	maxnorm=1.0
LRScheduler	ReduceLROnPlateau(factor=0.5,patience=5)
EarlyStopping	Patience=10(validationaccuracy)
Hardware	NVIDIARTX3050,6GBVRAM

Evaluationusesthreemetrics:MeanAbsoluteError(MAE,lowerisbetter),BinaryAccuracy,andWeightedF1Score(bothhigherisbetter).PearsonCorrelation(r)isadditionallyreported.

E.ExperimentalPhases

Phase1comparesthethreearchitectures(earlyfusion,latefusion, cross-attention with embeddim=64 and 4 heads) under identicalconditions.The64-dimbaselinewasintentional—itallowedtestingwhethercross-attentionisbeneficialatlowcapacitybeforecommittingtothelargerconfiguration.

Phase2isanablationstudy.Individualmodalitiesaredisabled:text-only,text+audio,text+vision,andthefullthree-modalitymodel.Allablationsusethesame64-dim,4-headconfiguration.

Phase 3 switches to embeddim=128 and hiddendim=128, thenvariesthenumberofattentionheads(8,16)anddropoutrate(0.1,0.5)relativetothedefaultPhase3configuration(Experiment3B).Thegoalwastofindtheconfigurationthatbestexploitstheincreasedcapacity.

*Note:*anearlierdraftofthisdescriptionreferencedathirdhead-countvalue(4)andathirddropoutvalue(0.2)thatarenot present in Table IV. Either report those two additional runs iftheyexistinthetraininglogs,orleavethetextasaboveiftheywerenevercompleted.

IV.RESULTSAND DISCUSSION

unimodalprocessingcanmatchcross-modalfusion.

D.TrainingSetup

Allmodelsweretrainedonalaptopwithan NVIDIA RTX3050GPU(6GBVRAM).Batchsizewassetto32

A.MainResults

Animportantobservationfrom Phase1pertainstothe lowest-performingconfiguration.Thecross-attentionmodel with4headsandembeddim=64(Experiment1C)achieved only 73.92%, which is lower than both baselines. This outcome

TABLEIV
ALL EXPERIMENTALRESULTSON CMU-MOSEITEST SET

Exp.	Description	MAE↓	Acc.↑	F1↑
1A	EarlyFusionBaseline	0.653	78.14	0.848
1B	LateFusionBaseline	1	%	0
1C	Cross-Attention-4H.	0.615	77.23	0.831
2A	Ablation:TextOnly	0.653	74.89	0.809
2B	Ablation: Text + Audio	5	%	2
2C	Ablation: Text + Vision	0.640	75.32	0.815
2D	Ablation:AllModali-ties	6	%	5
		0.627	76.65	0.830
3A	Proposed-8Heads	0.631	79.28	0.859
3	**Proposed-16Heads**	8	%	7
B	Dropout=0.1	**0.617**	**80.03**	**0.868**
3C	Dropout=0.5	**2**	**%**	**7**

highlightsthesensitivityofcross-attentiontorepresentational capacity. The initial assumption was that cross-attention should atleastmatchlatefusion,eveninaminimalconfiguration. Instead,itperformedroughlythreepointsbelowearlyfusion.

Theexplanationbecomesclearin Phase3.Withem-beddim=64and4heads,eachattentionheadoperatesina 16-dimensionalsubspace.Thatisnotenoughroomtolearn ausefulcross-modalrelationshipbetweena300-dimensional textembeddinganda74-dimensionalacousticfeaturevector. A possible explanation is that the model ends up with attention weightsthatarenearlyuniform,spreadingattentionevenly ratherthanfocusingonsemanticallymeaningfulrepresenta-tions.

Scaling to embeddim=128 and increasing head count fixed this. Eight heads produced79.28%. Sixteenheads pushed that to80.03%,thebestresultinthestudy.This6.11percentage pointgapbetweenthe4-headand16-headconfigurations, holdingeverythingelseconstant,isthecentralfindingofthe hyperparameteranalysis.

Earlyfusionreached78.14%.Thisiscompetitive,and notablybetterthantheunder-poweredcross-attentionmodel, whichreinforcesthepointthatapoorlyconfiguredTrans-formerisworsethanawell-implementedsimplebaseline.

Latefusionachieved77.23%accuracybutthelowestMAE ofanymodelat0.6150.Theindividualunimodalsub-models makefewercatastrophicallywrongregressionpredictions, buttheymissclassification-levelcross-modalsignals.Solate fusionproducesmoreconservative,lessextremeerrors,butis lessaccurateattheactualbinarypredictiontask.

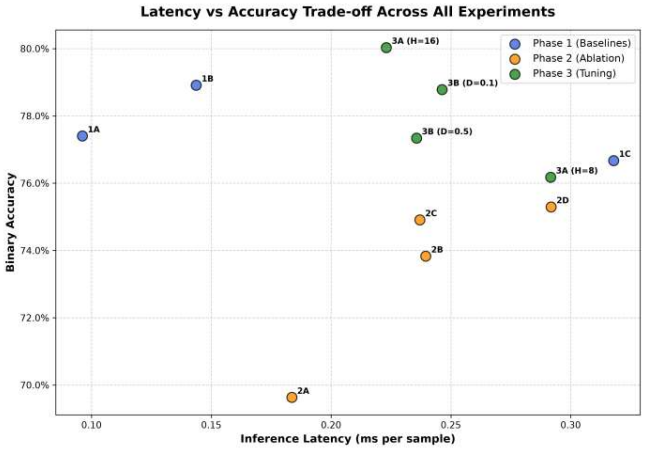
B.AblationStudy

TheablationresultsinTableIVshowthataddingau-dioorvisionmodalitiestoatext-onlybaselineconsistently improvesperformance—butonlywhenthearchitecturehas thecapacitytousethem.Text+Vision(76.65%)outperforms Text+Audio(75.32%),whichisabitsurprisinggiventhat YouTuberecordingscarrymorebackgroundnoise.Visionis audioisintuitivelymoreconnectedtovocalemotion.The likelyexplanationisthatFACETfacialactionunitfeatures

simplythecleanersignalinthisdataset.

The full three-modality ablation model (Text+Audio+Vision)at73.77%isactuallyworsethanText Onlyat74.89%.Thisisadirectconsequenceofinsufficient attention capacity—the 4-head, 64-dim model canno productivelyintegratethreeinputstreamssimultaneously,so addingmoremodalitiesintroducesnoiseratherthansignal.

*TODO(Aaron):*Experiment1CandExperiment2Dare nominallythesameconfiguration(4-head,embeddim=64, fullthree-modalitycross-attention)butreportdifferentnum- bers (73.92%/0.6407 MAE vs. 73.77%/0.6431 MAE). Confirm fromthetraininglogswhetherthesecamefromdifferent seeds/checkpoints,andeitherreconcilethenumbersoradd one sentence explaining the discrepancy (e.g. different random seed,differentearly-stoppingcheckpointselected).



(relatedtoOpenFace[25])arediscreteandrelativelyclean, whileCOVAREPacousticfeaturesextractedfromnaturalistic

Fig.2.Inferencelatency(mspersample)vs.binaryaccuracyforalleleven experimentsonCMU-MOSEI.

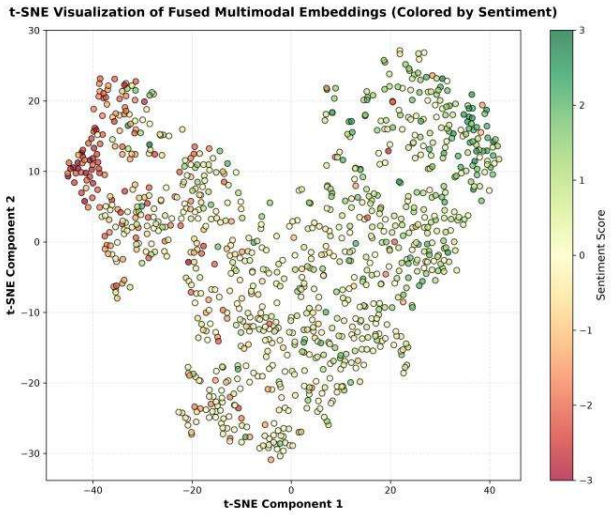


Fig. 3.t-SNE projection of fused multimodal embeddings from the best model

(16heads)

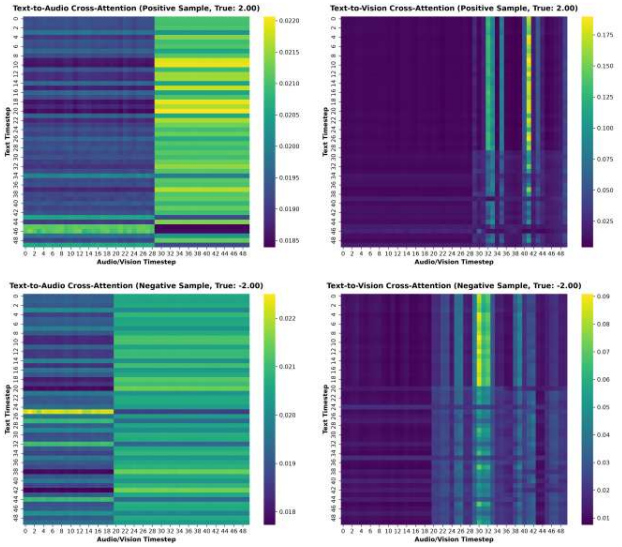


Fig.4. Cross-attentionheatmapsforapositiveutteranceandanegative utterance.

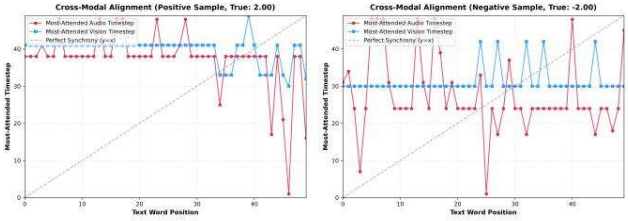


Fig.5.Cross-modalalignmentmaps.

C.AttentionAnalysis

The model heatmaps reveal an asymmetry between how the modelusesaudioversusvision.TheText-to-Visionpanels showstrong,narrowverticalbandsatasmallnumberof timesteps.Almosteverywordpositionattendsheavilyto those same frames, regardless of word position. The alignment makes this very clear: the vision mapping is nearly flat, sitting atoneortwotimestepvaluesacrossthefullsequence.

Thispatternsuggestsalearnedstrategy.Itimpliesthe modelhaslearnedtoidentifythemomentofpeakfacial expressionintheclipanduseitasastablereferencepoint fortheentireutterance,ratherthantryingtotrackword-by-face correspondence frame by frame. Given that CMU-MOSEI utterancesareshortclipswhereaspeakerisoftenholdinga sustainedexpressionthroughout,thisisactuallyareasonable strategy.

Audioattentionbehavesdifferently.Itoscillatessubstantiallyacrosswordpositions,sometimesjumpingtonearly oppositeendsoftheacousticsequencebetweenadjacent words.Theaudiosignalisbeingintegratedmorelocallyand

V.CONCLUSIONAND FUTURE WORK

ThispaperevaluatedaText-AnchoredCross-Attention Transformeragainstearlyandlatefusionbaselineson CMU-MOSEI. The main result—80.03% accuracy with 16 heads and embeddim=128—confirms that cross-attention fusion outperformssimpleralternativeswhengivensufficientrepresentationalcapacity.Themoreinterestingresultisthatitdoesnot outperformthemwithinsufficientcapacity;afour-head,64-dimmodelscoredbelowbothbaselinesat73.92%.

Theprimarylimitationofthisworkistheuseof300-dimGloVestatictextembeddings.Publishedsystemsachieving above82%onCMU-MOSEIconsistentlyuseBERTor RoBERTacontextualembeddings.Thepre-extractedCMU-MOSEIfeaturesusedheredonotincludeBERTrepresentations,soswitchingwouldrequireeitherfindingasepa-ratelyextractedBERTfeaturesfileorrunningextractionfrom scratch—aworthwhilenextstepthatwasoutsidethescopeof thisphase.

Other planned extensions include replacing FACET features withajointlytrainedVisionTransformerencoder,evaluating on IEMOCAP [23] and MER 2024 [24], as well as future challenges likeMER2025[26]forcross-datasetgeneralization, variably.Whetherthisismeaningfulacousticsensitivityor partialnoise-chasingishardertosaywithoutaccesstothe rawaudio.

andexploringCLIP-basedvisualfeaturestoleveragevision-languagepretraininginthecross-attentionmechanism.

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